

Exercise 1

Arham Lodha

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Problem 1.

Proof. $\forall z \in H, \Im(z) > 0 \implies \Im(z+i) > 0 \implies z+i \neq 0 \implies f$ is well defined for all $z \in H$.

$\forall z \in D, |z| < 1 \implies z \neq 1 \implies 1-z \neq 0 \implies g$ is well defined for $z \in D$.

$\forall z \in H,$

$$\begin{aligned} g(f(z)) &= g\left(\frac{z-i}{z+i}\right) \\ &= i \left(\frac{\frac{z-i}{z+i} + 1}{1 + \frac{i-z}{z+i}} \right) \\ &= i \left(\frac{z-i+z+i}{z+i+i-z} \right) \\ &= \frac{2zi}{2i} = z \end{aligned}$$

$\forall w \in D,$

$$\begin{aligned} f(g(w)) &= f\left(i \frac{w+1}{1-w}\right) \\ &= \frac{i\left(\frac{w+1}{1-w} - 1\right)}{i\left(\frac{w+1}{1-w} + 1\right)} \\ &= \frac{w+1-1+w}{w+1+1-w} \\ &= \frac{2w}{2} = w \end{aligned}$$

Thus $g \circ f = \text{id}_H$ and $f \circ g = \text{id}_D$. Thus g and f are reciprocal conformal equivalences □

Problem 2a.

Proof. $\forall a \in K,$

$$\begin{bmatrix} a & \\ & a^{-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ a-1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -a^{-1} \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ a-a^2 & 1 \end{bmatrix}$$

□

Problem 2b.

Proof. $\forall A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL_2(\mathbb{R})$,

$$\begin{aligned} \begin{bmatrix} 1 & \\ -\frac{c}{a} & 1 \end{bmatrix} A \begin{bmatrix} a^{-1} & \\ & a \end{bmatrix} \begin{bmatrix} 1 & -ab \\ & 1 \end{bmatrix} &= \begin{bmatrix} 1 & \\ -\frac{c}{a} & 1 \end{bmatrix} \begin{bmatrix} 1 & ba \\ \frac{c}{a} & da \end{bmatrix} \begin{bmatrix} 1 & -ab \\ & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & ba \\ 0 & da - bc \end{bmatrix} \begin{bmatrix} 1 & -ab \\ & 1 \end{bmatrix} = \begin{bmatrix} 1 & ba \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & -ab \\ & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Thus,

$$A = \begin{bmatrix} 1 & \\ \frac{c}{a} & 1 \end{bmatrix} \begin{bmatrix} 1 & ab \\ & 1 \end{bmatrix} \begin{bmatrix} a & \\ & a^{-1} \end{bmatrix}$$

□

Problem 2c.

Proof. Let $A(n) = \begin{bmatrix} 1 & n \\ & 1 \end{bmatrix}$ and $B(n) = \begin{bmatrix} 1 & \\ n & 1 \end{bmatrix}$

For $i = 1$:

$$\begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}^1 = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & \\ 1 & 1 \end{bmatrix}^1 = \begin{bmatrix} 1 & \\ 1 & 1 \end{bmatrix}$$

Suppose for $n \in \mathbb{N}$: $A(1)^n = A(n)$ and $B(1)^n = B(n)$.

$$A(1)^{n+1} = A(1)A(n) = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & n \\ & 1 \end{bmatrix} = \begin{bmatrix} 1 & n+1 \\ & 1 \end{bmatrix} = A(n+1)$$

$$B(1)^{n+1} = B(1)B(n) = \begin{bmatrix} 1 & \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ n & 1 \end{bmatrix} = \begin{bmatrix} 1 & \\ n+1 & 1 \end{bmatrix} = B(n+1)$$

Thus $A(1)$ and $B(1)$ generate $SL_2(\mathbb{F}_p)$ by part a and b.

□

Problem 2d.

Proof. $\forall z \in \mathbb{Z}$, let $\bar{z}$ denote the equivalence class mod p . Note the function $n \rightarrow \bar{n}$ is a ring homomorphism thus $n + m \rightarrow \overline{n + m} = \bar{n} + \bar{m}$ and $nm \rightarrow \overline{nm} = \bar{n} * \bar{m}$. $\forall A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL_2(\mathbb{Z})$, let $\bar{A} \in SL_2(\mathbb{F}_p)$

be $\begin{bmatrix} \bar{a} & \bar{b} \\ \bar{c} & \bar{d} \end{bmatrix}$, note $\overline{ad - bc} = \bar{ad} - \bar{bc} = \bar{1}$.

Let $q : SL_2(\mathbb{Z}) \rightarrow SL_2(\mathbb{F}_p)$ where $A \rightarrow \bar{A}$.

$$\begin{aligned}
q\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix}\right) &= q\left(\begin{bmatrix} aw + by & ax + bz \\ cw + dy & cx + dz \end{bmatrix}\right) \\
&= \begin{bmatrix} \overline{aw + by} & \overline{ax + bz} \\ \overline{cw + dy} & \overline{cx + dz} \end{bmatrix} \\
&= \begin{bmatrix} \overline{aw} + \overline{by} & \overline{ax} + \overline{bz} \\ \overline{cw} + \overline{dy} & \overline{cx} + \overline{dz} \end{bmatrix} \\
&= \begin{bmatrix} \overline{a} & \overline{b} \\ \overline{c} & \overline{d} \end{bmatrix} \begin{bmatrix} \overline{w} & \overline{x} \\ \overline{y} & \overline{z} \end{bmatrix}
\end{aligned}$$

Thus q is a homomorphism of groups. Moreover note $q\left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right) = \begin{bmatrix} \overline{1} & \overline{1} \\ \overline{1} & \overline{1} \end{bmatrix}$ and $q\left(\begin{bmatrix} 1 & \\ 1 & 1 \end{bmatrix}\right) = \begin{bmatrix} \overline{1} & \\ \overline{1} & \overline{1} \end{bmatrix}$ and as previously proven, $\begin{bmatrix} \overline{1} & \overline{1} \\ \overline{1} & \overline{1} \end{bmatrix}$ and $\begin{bmatrix} \overline{1} & \\ \overline{1} & \overline{1} \end{bmatrix}$ generate $SL_2(\mathbb{F}_p)$. Thus q is surjective. $\square$

Problem 2e.

Proof. $\forall q \geq 1$: The natural projection is $\pi : SL_2(\mathbb{Z}) \rightarrow SL_2(\mathbb{Z}/q\mathbb{Z})$ is a group homomorphism with $\Gamma(q)$ being the kernel. Thus since $\mathbb{Z}/q\mathbb{Z}$ is finite $SL_2(\mathbb{Z}/q\mathbb{Z})$ is finite and since $SL_2(\mathbb{Z})/\Gamma(q) \cong SL_2(\mathbb{Z}/q\mathbb{Z}) \implies |SL_2(\mathbb{Z})/\Gamma(q)| = |SL_2(\mathbb{Z}/q\mathbb{Z})| < \infty$. Note $\Gamma(q) < \Gamma_1(q) < \Gamma_0(q) < SL_2(\mathbb{Z}) \implies \infty > |SL_2(\mathbb{Z})/\Gamma(q)| > |SL_2(\mathbb{Z})/\Gamma_1(q)| > |SL_2(\mathbb{Z})/\Gamma_0(q)|$

For p prime: $SL_2(\mathbb{F}_p) = \ker \det$. $|GL_2(\mathbb{F}_p)| = (p^2 - 1)(p^2 - p) = p(p^2 - 1)(p - 1)$. Thus $|SL_2(\mathbb{F}_p)| = |GL_2(\mathbb{F}_p)| / |\mathbb{F}_p^\times| = p(p^2 - 1)$. Thus $|SL_2(\mathbb{Z}) : \Gamma(p)| = p(p^2 - 1)$.

Let $\pi : \Gamma_0(p) \rightarrow G$ where G is upper triangular matrices in $SL_2(\mathbb{F}_p)$. Note G is a group and $\ker \pi = \Gamma(p)$. Thus $|G| = [\Gamma_0(p) : \Gamma(p)]$. $\forall A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \in M_{2 \times 2}(\mathbb{F}_p), A \in G \iff a \cdot c = 1$. So there are $p - 1$ choices for a and p choices for b . Thus $[\Gamma_0(p) : \Gamma(p)] = |G| = p(p - 1)$. Thus $[SL_2(\mathbb{Z}) : \Gamma_0(p)] = [SL_2(\mathbb{Z}) : \Gamma(p)] / [\Gamma_0(p) : \Gamma(p)] = p + 1$.

Let $\pi : \Gamma_1(p) \rightarrow H$ where H is the subgroup of matrices of form $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ in $SL_2(\mathbb{F}_p)$. $[\Gamma_1(p) : \Gamma(p)] = |H| = p$. Thus $[SL_2(p) : \Gamma_1(p)] = [SL_2(p) : \Gamma(p)] / [\Gamma_1(p) : \Gamma(p)] = p^2 - 1$. $\square$

Problem 3a.

Proof. $\forall f \in C(X), \forall g, h \in G$ and $\forall x \in X$.

$$\begin{aligned}
((f \cdot g) \cdot h)(x) &= (f \cdot g)(h \cdot x) \\
&= f(gh \cdot x) \\
&= (f \cdot gh)(x)
\end{aligned}$$

$\square$

Problem 3b.

Proof. Let $\alpha : G \times X \rightarrow \mathbb{C}^\times$. $\implies$: Suppose $\forall f \in C(X), \forall g \in G, \forall x \in X$,

$$(f \cdot g)(x) = \alpha(g, x)f(g \cdot x)$$

defines an action of G on $C(X)$.

$\forall g, h \in G$

$$\alpha(gh, x)f(gh \cdot x) = (f \cdot gh)(x) = ((f \cdot g) \cdot h)(x) = \alpha(h, x)(f \cdot g)(h \cdot x) = \alpha(h, x)\alpha(g, h \cdot x)f(gh \cdot x)$$

Let $f : X \rightarrow \mathbb{C}$ where $x \rightarrow 1$.

Thus

$$\alpha(gh, x) = \alpha(h, x)\alpha(g, h \cdot x)$$

$\Leftarrow$: Suppose $\forall g, h \in G$

$$\alpha(gh, x) = \alpha(h, x)\alpha(g, h \cdot x)$$

Then $\forall f \in C(X), \forall x \in X, \forall g, h \in G$

$$\begin{aligned} ((f \cdot g) \cdot h)(x) &= \alpha(h, x)(f \cdot g)(hx) \\ &= \alpha(h, x)\alpha(g, hx)f(ghx) \\ &= \alpha(gh, x)f(ghx) \\ &= (f \cdot gh)(x) \end{aligned}$$

Thus $(f \cdot g)(x) = \alpha(g, x)f(gx)$ is a valid action of G on $C(X)$. □

Problem 3c.

Proof. Let $\gamma : X \rightarrow \mathbb{C}^\times$ be any function. Let $\alpha : G \times X \rightarrow \mathbb{C}^\times$ where $\forall (g, x) \in G \times X$

$$\alpha(g, x) = \frac{\gamma(g \cdot x)}{\gamma(x)}$$

$\forall g, h \in G$ and $\forall x \in X$

$$\begin{aligned} \alpha(h, x)\alpha(g, hx) &= \frac{\gamma(h \cdot x)}{\gamma(x)} \frac{\gamma(gh \cdot x)}{\gamma(h \cdot x)} \\ &= \frac{\gamma(gh \cdot x)}{\gamma(x)} \\ &= \alpha(gh, x) \end{aligned}$$

Thus α has the property seen in 3b. □

Problem 4a.

Proof. Let $x_0 \in X$ be fixed. Let $\lambda : V_H \rightarrow C(G)$ be a linear map defined $\lambda(f) = [g \rightarrow f(g \cdot x_0)\alpha(g, x_0)]$.

$\forall k \in \ker \lambda, \lambda(k) = [g \rightarrow 0]. \forall g \in G, k(g \cdot x_0)\alpha(g, x_0) = 0$. Since $\alpha \in \text{Hom}(G \times X, \mathbb{C}^\times) \implies \alpha(g, x_0) \in \mathbb{C}^\times \implies \alpha(g, x_0) \neq 0$. Thus $k(g \cdot x_0) = 0$ for all $g \in G$. Thus $(k \cdot g)(x_0) = 0$. Since $k \in V_H$, $(k \cdot g)(x_0) = k(x_0) = 0$. Since G acts transitively on X , $\forall y \in X, \exists g \in G \ni g \cdot x_0 = y \implies k(y) = k(g \cdot x_0) = (k \cdot g)(x_0) = k(x_0) = 0$. Thus $k = 0$. Thus $\ker \lambda = 0 \implies \lambda$ is injective. $\square$

Problem 4b.

Proof. $\forall f \in V_H$, let $\tilde{f} = \lambda(f)$. $\forall (g, h) \in G \times H$

$$\begin{aligned} \tilde{f}(hg) &= f(hg \cdot x_0)\alpha(hg, x_0) \\ &= f(hg \cdot x_0)\alpha(g, x_0)\alpha(h, gx_0) \\ &= (f \cdot h)(g \cdot x_0)\alpha(g, x_0) \\ &= f(g \cdot x_0)\alpha(g, x_0) \\ &= \tilde{f}(g) \end{aligned}$$

$\square$

Problem 4c.

Proof. Let $K = G^{x_0}$. Let $\chi : K \rightarrow \mathbb{C}^\times$ where $k \rightarrow \alpha(k, x_0)$. $\forall g, h \in K$,

$$\begin{aligned} \chi(gh) &= \alpha(gh, x_0) \\ &= \alpha(h, x_0)\alpha(g, h \cdot x_0) \\ &= \alpha(h, x_0)\alpha(g, x_0) \\ &= \chi(h)\chi(g) \end{aligned}$$

Thus χ is a group homomorphism. $\square$

Problem 4d.

Proof. $\forall f \in V_H, \forall k \in K$ and $g \in G$

$$\begin{aligned} \tilde{f}(gk) &= f(gk \cdot x_0)\alpha(gk, x_0) \\ &= f(g \cdot x_0)\alpha(k, x_0)\alpha(g, kx_0) \\ &= f(g \cdot x_0)\alpha(g, x_0)\alpha(k, x_0) \\ &= \tilde{f}(g)\chi(k) \end{aligned}$$

$\square$

Problem 4e.

Proof. Suppose $\tilde{f} : G \rightarrow \mathbb{C}$ is a function such that $\forall h \in H, \forall k \in K, \forall g \in G \tilde{f}(hg) = \tilde{f}(g)$ and $\tilde{f}(gk) = \chi(k)\tilde{f}(g)$.

Since G acts transitively on X , then $\forall x \in X \exists g_x \in G \ni g_x \cdot x_0 = x$. Define $f(x) = \frac{\tilde{f}(g_x)}{\alpha(g_x, x_0)}$.

Well definedness: Suppose $\exists x \in X \ni \exists g, g' \in G \ni x = g \cdot x_0 = g' \cdot x_0 \implies g = g'k$ for some $k \in K$.
Thus

$$\begin{aligned} f(x) &= \frac{\tilde{f}(g)}{\alpha(g, x_0)} \\ &= \frac{\tilde{f}(g'k)}{\alpha(g'k, x_0)} \\ &= \frac{\tilde{f}(g')\chi(k)}{\alpha(g', k \cdot x_0)\chi(k)} \\ &= \frac{\tilde{f}(g')}{\alpha(g', x_0)} \end{aligned}$$

so f is well defined.
 $\forall x \in X$ and $\forall h \in H$

$$\begin{aligned} (f * h)(x) &= \alpha(h, x)f(h \cdot x) \\ &= \alpha(h, x) \frac{\tilde{f}(hg)}{\alpha(hg, x_0)} \\ &= \alpha(h, x) \frac{\tilde{f}(hg)}{\alpha(g, x_0)\alpha(h, gx_0)} \\ &= \frac{\tilde{f}(g)}{\alpha(g, x_0)} \\ &= f(x) \end{aligned}$$

Finally $\forall g \in G$,

$$\lambda(f)(g) = f(gx_0)\alpha(g, x_0) = \tilde{f}(g)$$

Thus image of λ is the set of functions $\tilde{f} : G \rightarrow \mathbb{C}$ such that (2) and (3) are valid. $\square$

Problem 5a.

Proof. $\forall k \in \mathbb{Z}$. $\forall A_1, A_2 \in SL_2(\mathbb{R})$ where $A_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$ and $A_2 = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$.

$$\text{Note } A_1 A_2 = \begin{bmatrix} a_1 a_2 + b_1 c_2 & a_1 d_2 + b_1 d_2 \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{bmatrix}.$$

Thus

$$\begin{aligned}
\alpha(A_1 A_2, z) &= ((c_1 a_2 + d_1 c_2)z + c_1 b_2 + d_1 d_2)^k \\
&= (c_1(a_2 z + b_2) + d_1(c_2 z + d_2))^k \\
&= (c_2 z + d_2)^k \left(\frac{c_1(a_2 z + b_2) + d_1(c_2 z + d_2)}{c_2 z + d_2} \right)^k \\
&= (c_2 z + d_2)^k \left(c_1 \frac{a_2 z + b_2}{c_2 z + d_2} + d_1 \right) \\
&= \alpha(A_2, z) \alpha(A_1, A_2 \cdot z)
\end{aligned}$$

For a modular for f of weight k it follows the following rule $f\left(\frac{az+b}{cz+d}\right)(cz+d)^k f(z)$ for $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \Gamma \in SL_2(\mathbb{Z})$. Thus α is the multiplication factor. $\square$

Problem 5b.

Proof. Assume the contrary that for $k \neq 0$, $\exists \lambda \in Hom(H, \mathbb{C}^x) \ni \alpha(g, x) = \frac{\lambda(gx)}{\lambda(x)}$. Then $\alpha(-I, z) = (-1)^k$ and $\alpha(-I, z) = \frac{\lambda(z)}{\lambda(z)} = 1$. Thus $(-1)^k = 1$. Thus we have 2 cases:

1. k is odd: Immediate contradiction

2. k is even: $\forall r_\theta = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \in SO(2)$. $\alpha(r_\theta, i) = (-\sin(\theta)i + \cos(\theta))^k = e^{-ki\theta}$. Also $\alpha(r_\theta, i) = \lambda(r_\theta, i)/\alpha(i) = \alpha(i)/\alpha(i) = 1$. Thus $\forall \theta$, $e^{-ki\theta} = 1$. This is a contradiction because if $\theta = \frac{\pi}{k} \implies e^{-ki\frac{\pi}{k}} = e^{-i\pi} = -1 \neq 1$.

Thus by contradiction, α is not of the form shown in 3c. $\square$